MATHEMATICS

Analysis and approaches

Higher Level | Set A | Paper 2

Mock examination · 2025 series

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A graphic display calculator is required.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 5]

The mass X grams of an item is modelled by a normal distribution with mean 230 and standard deviation 9.

(a) Find $\mathrm{P}(X > 240)$. [2]

(b) Find a such that $\mathrm{P}(a < X < 240) = 0.5$. [3]

SECTION A

Question 2 [Maximum mark: 5]

The function f is defined by $f(x) = x^2 \ln(x-2) + 1$, $x > 2$.

(a) Find $f\prime(3)$. [2]

(b)(i)) Find the equation of the normal to $y = f(x)$ at $x = 3$, in point-gradient form. [2]

(b)(ii)) Write this normal equation in the form $y = mx + c$. [1]

SECTION A

Question 3 [Maximum mark: 5]

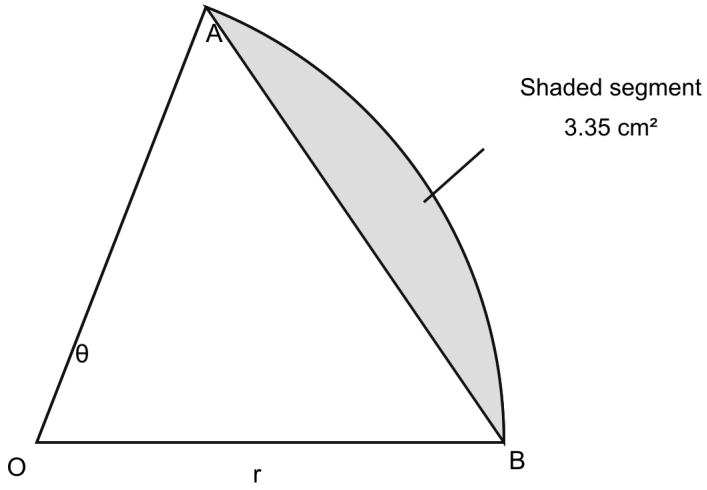
In the binomial expansion of $(2x + ax^2)^7$, the coefficient of x^9 is 6048, where $a \in \mathbb{R}$.

Find all possible values of a. [5]

SECTION A

Question 4 [Maximum mark: 6]

A minor sector OAB has radius r cm and angle θ radians, where $0 < \theta < \pi$. Its perimeter, including the two radii, is 16 cm. The area of the segment bounded by chord AB and minor arc AB is 3.35 cm^2.



Sector perimeter 16 cm; diagram not to scale

Find r and θ, giving each answer to three significant figures. [6]

SECTION A

Question 5 [Maximum mark: 8]

For $n \in \mathbb{Z}^+$, define $u_n = \dfrac{12^{n+1} + 2 \cdot 5^n}{14}$.

(a) Prove by mathematical induction that u_n is an integer for every $n \in \mathbb{Z}^+$. [6]

(b) Find the smallest positive integer n for which $u_n > 10^6$. [2]

SECTION A

Question 6 [Maximum mark: 8]

This question concerns sketches from qualitative information. Numerical coordinates of any intercept not specified need not be calculated. Draw each sketch on a separate set of axes.

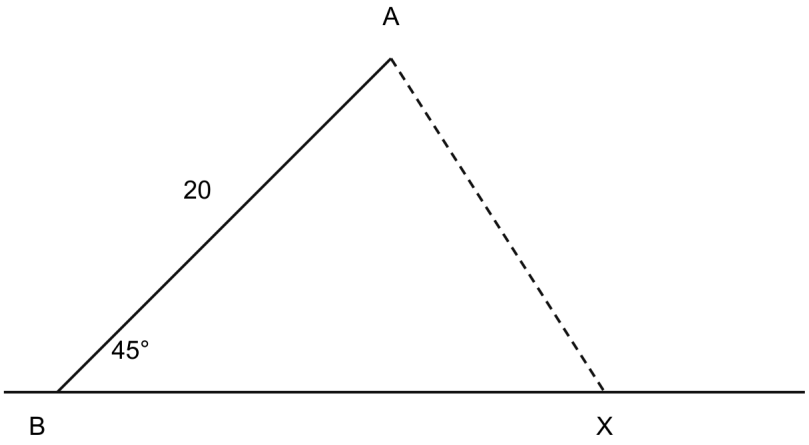
(a) The differentiable function f is even and tends to 3 as $x \to \pm\infty$. It satisfies $f(0) = -3$, $f(\pm 1) = 1$ and $f(\pm 2) = 4$. Its derivative is positive for $x < -2$ and $0 < x < 2$, negative for $-2 < x < 0$ and $x > 2$, and zero only at $-2, 0, 2$. Sketch a possible graph of $y = f(x)$, showing its stationary points and horizontal asymptote. [4]

(b) The function g is differentiable for $x \neq 1$. Its limits are $g(x) \to -2$ as $x \to -\infty$, $g(x) \to 1$ as $x \to +\infty$, $g(x) \to -\infty$ as $x \to 1^-$ and $g(x) \to +\infty$ as $x \to 1^+$. It has $g\prime(x) > 0$ for $x < 0$ and $x > 2$, and $g\prime(x) < 0$ for $0 < x < 1$ and $1 < x < 2$. The only stationary points are at $x = 0$ and $x = 2$, with $-1 < g(0) < 0$ and $0 < g(2) < 1$. Sketch a possible graph, showing all asymptotes and stationary points. The exact stationary ordinates are not required. [4]

SECTION A

Question 7 [Maximum mark: 6]

Points A and B are fixed with $AB = 20$. A ray starts at B and makes an angle of $45°$ with BA. The point X may lie anywhere on this ray other than B.



X may move along the ray; diagram not to scale

(a) Find the exact value of BX when AX is as small as possible. [2]

(b) Find the exact value of BX when $AX = 20$. [1]

(c) When $AX = 17$, find both possible values of BX. [3]

SECTION A

Question 8 [Maximum mark: 6]

Let $w = 2 + 3i$. A polynomial is defined by $P(z) = z^4 + az^2 + 72z + b$, where $a, b \in \mathbb{R}$. It is given that $P(w) = 0$.

(a) Find w^4 in the form $u + vi$. [1]

(b) Find a and b. [4]

(c) Find the other three zeros of P. [1]

SECTION A

Question 9 [Maximum mark: 6]

The function $\arctan$ takes its principal values in $(-\pi/2, \pi/2)$. Angles are in radians.

(a) Give a counterexample to the claim that $\arctan(\tan x) = x$ for every real x for which $\tan x$ exists. [2]

(b) Prove that $\arctan x + \arctan(1/x) = \pi/2$ for every $x > 0$. [4]

SECTION B

Question 10 [Maximum mark: 17]

The line l_1 has Cartesian equation $\frac{x+16}{-5} = \frac{y+4}{-7} = z$. The line l_2 passes through $A(3, 4, 1)$ and $B(0, 6, 0)$. Let $C = (-11, 3, -1)$.

(a) Find a vector equation for l_2. [2]

(b) Show that l_1 and l_2 meet, and find their point of intersection P. [5]

(c) Show that the lines are perpendicular. [2]

(d) Show that C lies on l_1. [2]

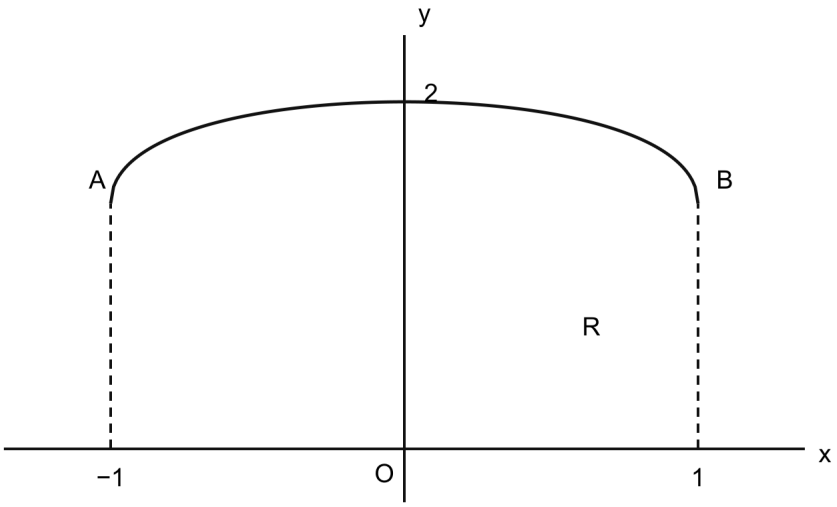
(e) The point D is the reflection of C in the line l_2. Find D. [2]

(f) Find a Cartesian equation of the plane containing both lines. [4]

SECTION B

Question 11 [Maximum mark: 19]

Let $f(x) = \sqrt{1+x} + \sqrt{1-x}$, $-1 \leq x \leq 1$. Its graph has endpoints A at $x = -1$ and B at $x = 1$. Let R be the region bounded by this graph, the two coordinate axes and the line $x = 1$, in the first quadrant.



(a)(i)) Find the exact y-coordinate of A. [1]

(a)(ii)) State the range of f. [2]

(b) The horizontal line $y = k$ is tangent to the graph. Find k. [1]

(c) The line OB, where O is the origin, has gradient m. Find m. [1]

(d) Find all real k for which the line $y = kx$ does not meet the graph. [3]

(e) Find the exact area of R. [4]

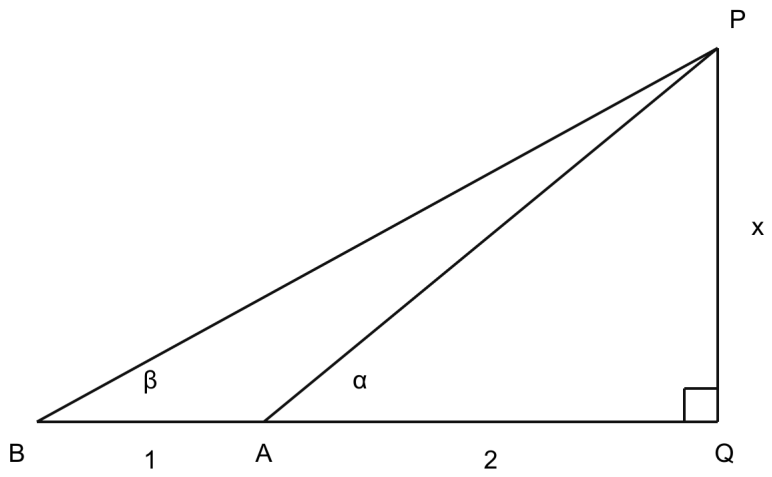
(f) Find the volume generated when R is rotated through 2π radians about the x-axis. [2]

(g) Find the volume generated when R is instead rotated through 2π radians about the y-axis. Show your method. [5]

SECTION B

Question 12 [Maximum mark: 19]

Points B, A, Q lie in this order on a horizontal line, with $BA = 1$ and $AQ = 2$. The point P is vertically above Q with $QP = x > 0$. Let $\alpha = \angle PAQ$ and $\beta = \angle PBQ$. Define $f(x) = x/(6 + x^2)$ for $x > 0$.



Schematic; all lengths in the same units

(a) Express α and β in terms of x. [2]

(b) Show that $\alpha - \beta = \arctan\left(\frac{x}{6 + x^2}\right)$. [3]

(c) Find the exact coordinates of the maximum point of $y = f(x)$, justifying your answer. [5]

(d) The graph $y = f(x)$ has a point of inflexion at $x = p$, where $p > 0$. Find p exactly. [4]

(e)(i)) Find the maximum value of $\angle APB$ in degrees. [2]

(e)(ii)) Find both possible values of x for which $\angle APB = 10°$. [3]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 5]

(a) $\mathrm{P}(X > 240) = 0.13326... \cong 0.133$. **M1 A1**

[2 marks]

(b) $\mathrm{P}(X > a) = 0.5 + 0.13326... = 0.63326...$ **M1 A1**

Hence $a = 226.9355 \cong 227$ **A1**

[3 marks]

2. [Maximum mark: 5]

Let $f(x) = x^2 \ln(x-2) + 1$

(a) By GDC,

$f'(3) = 9$ **A2**

[2 marks]

(b) By GDC,

(i) $y - 1 = -\frac{1}{9}(x-3)$; **A1 A1**

(ii) $y = -0.111x + 1.33$ **OR** $y = -\frac{1}{9}x + \frac{4}{3}$ **A1**

[3 marks]

Remark

GDC gives all the results.

You can find the result analytically, but it will be more time consuming and risky!

For $x = 3$, $y = 9\ln 1 + 1 = 1$, so the point is (3,1)

$$f'(x) = 2x\ln(x-2) + \frac{x^2}{x-2}$$

$$f'(3) = 6\ln 1 + \frac{9}{1} = 9$$

and the results follow!

3. [Maximum mark: 5]

$^7C_r\left(2x\right)^{7-r}\left(ax^2\right)^r$ **M1**

The power of x is $7-r+2r=7+r$

$7+r=9 \Leftrightarrow r=2$ **A1**

Hence, the coefficient is $^7C_2\left(2x\right)^5\left(a\right)^2=672a^2$ **A1**

$672a^2=6048$ **M1**

$a^2=9 \Leftrightarrow a=\pm 3$ **A1**

[5 marks]

4. [Maximum mark: 6]

$\frac{1}{2}r^2(\theta-\sin\theta)=3.35 \Leftrightarrow r^2(\theta-\sin\theta)=6.7$ **M1 A1**

$r\theta+2r=16 \Leftrightarrow r(\theta+2)=16 \Leftrightarrow r=\frac{16}{\theta+2}$ **A1**

$\left(\frac{16}{\theta+2}\right)^2(\theta-\sin\theta)=6.7$ **M1**

$\theta=1.2$ **A1**

$r=5$ **A1**

[6 marks]

5. [Maximum mark: 8]

(a) For $n=1$, $12^2+2\times5^1=154$, which is divisible by 14. (since $154=11\times14$). **A1**

Assume it is true for $n=k$, i.e.

$12^{k+1}+2\times5^k=14m$, for some $m\in\mathbf{Z}^+$. **M1**

Prove it is true for $n=k+1$, i.e.

$12^{k+2}+2\times5^{k+1}=14m'$ for some $m'\in\mathbf{Z}^+$. **M1**

Indeed,

$12^{k+2}+2\times5^{k+1}=12\times12^{k+1}+10\times5^k$

$=12\times(14m-2\times5^k)+10\times5^k$ **A1**

$=12\times14m-24\times5^k+10\times5^k$

$=12\times14m-14\times5^k$

$=14(12m-5^k)$ which is divisible by 14. **A1**

The statement is true for $n=1$, and assuming it is true for $n=k$, we proved it is also true for $n=k+1$. Therefore, by mathematical induction it is true for any $n\in\mathbf{Z}^+$. **R1**

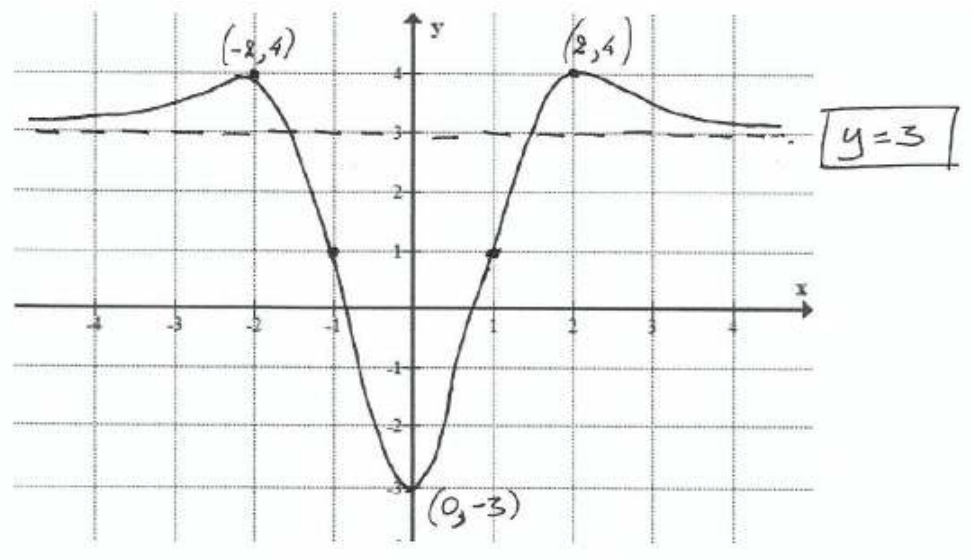
[6 marks]

(b) For $n=6$, the integer $\dfrac{12^7+2\times5^6}{14}=2561647$ which exceeds 10^6. **A1 A1**

[2 marks]

6. [Maximum mark: 8]

(a)

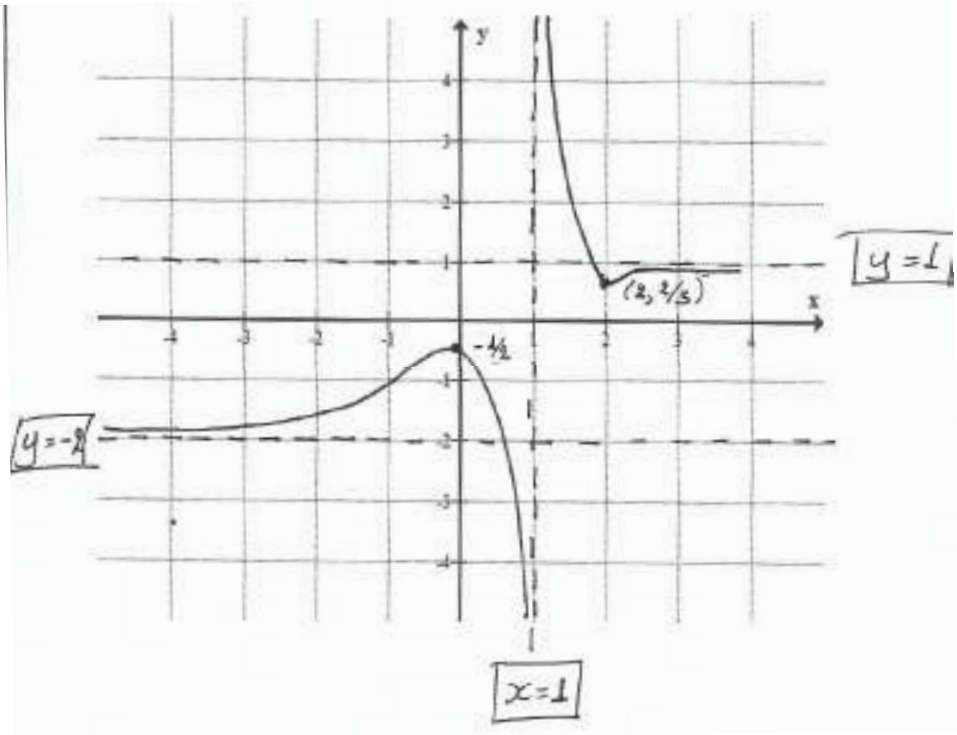


Asymptote, symmetric graph **A1 A1**

Two local max and a local min **A1 A1**

[4 marks]

(b)

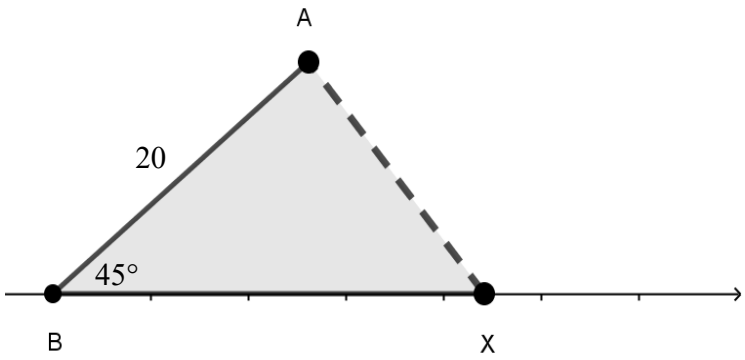


Asymptotes and graph **A1 A1**

One local maximum and one local minimum **A1 A1**

[4 marks]

7. [Maximum mark: 6]



(a) $\cos 45° = \frac{BX}{20} \Rightarrow BX = 10\sqrt{2}$ **M1 A1**

[2 marks]

(b) $BX = 2 \times 10\sqrt{2} = 20\sqrt{2}$ **A1**

[1 mark]

(c) $17^2 = BX^2 + 20^2 - 2 \times 20 \times BX \cos 45°$. **M1**

$BX = \sqrt{89} + 10\sqrt{2} \cong 23.6$ **A1**

$BX = -\sqrt{89} + 10\sqrt{2} \cong 4.71$ **A1**

[3 marks]

Remark

For 7(c), you can also use sine rule to find the two triangles in the ambiguous case, although it is more time consuming.

8. [Maximum mark: 6]

(a) By GDC, $w^4 = -119 - 120\,\mathrm{i}$. **A1**

[1 mark]

(b) Also by GDC, $w^2 = -5 + 12\,\mathrm{i}$. **A1**

$P(w) = w^4 + aw^2 + 72w + b = 0$ **M1**

$(-119 - 120\,\mathrm{i}) + a(-5 + 12\mathrm{i}) + 72(2 + 3\mathrm{i})) + b = 0$

Real Part: $-119 - 5a + 144 + b = 0 \Leftrightarrow 5a - b = 25$ **A1**

Imaginary Part: $-120 + 12a + 216 = 0$

Hence, $a = -8$, $b = -65$ **A1**

[4 marks]

(c) For $P(z) = z^4 - 8z^2 + 72z - 65$, the GDC gives the zeros

1, -5, $2 - 3\mathrm{i}$ (and $2 + 3\mathrm{i}$) **A1**

[1 mark]

9. [Maximum mark: 6]

(a) For $x = \frac{3\pi}{4}$ **A1**

LHS = $\arctan\left(\tan\frac{3\pi}{4}\right) = \arctan(-1) = -\frac{\pi}{4}$ RHS = $\frac{3\pi}{4}$ **R1**

[2 marks]

(b) Let $A = \arctan x$ and $B = \arctan\frac{1}{x}$

$\tan(A+B) = \frac{x+\frac{1}{x}}{1-1}$, not defined. **M1 A2**

Hence, since both A and B are positive, $A+B = \frac{\pi}{2}$. **R1**

[The right-angled triangle method could also be used]

[4 marks]

Section B

10. [Maximum mark: 17]

(a) $l_2: \quad r = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} + t \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix}$ **A1 A1**

[2 marks]

(b) $l_1: \quad r = \begin{pmatrix} -16 \\ -4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ -7 \\ 1 \end{pmatrix}$ **A1**

$$\begin{aligned} 3 - 3t &= -16 - 5\lambda \\ 4 + 2t &= -4 - 7\lambda \\ 1 - t &= \lambda \end{aligned} \quad \Leftrightarrow \quad \begin{aligned} 3t - 5\lambda &= 19 \\ 2t + 7\lambda &= -8 \\ t + \lambda &= 1 \end{aligned}$$ **M1**

The first two equations (or any two equations) give: $t = 3$ and $\lambda = -2$ **A1**

which satisfy the third equation, so the two lines intersect. **R1**

The position vector of their point of intersection is $r = \begin{pmatrix} -6 \\ 10 \\ -2 \end{pmatrix}$ **A1**

[5 marks]

(c) $\begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix} \bullet \begin{pmatrix} -5 \\ -7 \\ 1 \end{pmatrix} = 15 - 14 - 1 = 0$ **M1 A1**

[2 marks]

(d) For $\lambda = -1$, the equation of the line gives C. **M1 A1**

[2 marks]

(e) C(-11, 3, -1) P(-6, 10, -2) D ?

P is the midpoint of CD **M1**

Hence, D(-1, 17, -3) **A1**

[2 marks]

(f) $\begin{pmatrix}-3\\2\\-1\end{pmatrix}\times\begin{pmatrix}-5\\-7\\1\end{pmatrix}=\begin{pmatrix}-5\\8\\31\end{pmatrix}$ **M1 A1**

The line passes through (3,4,1) [(OR (-16,-4,0)]

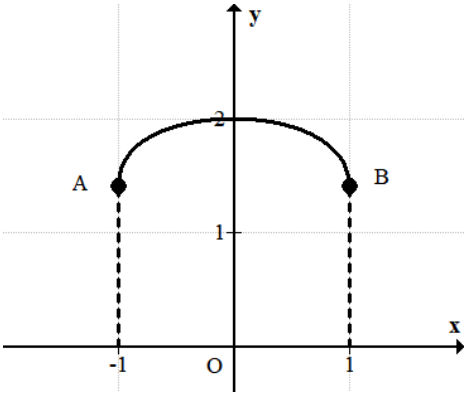
Hence

$-5x+8y+31z=48$ **M1 A1**

[4 marks]

11. [Maximum mark: 19]

$f(x) = \sqrt{1+x} + \sqrt{1-x}$, $-1 \le x \le 1$.



(a) (i) $\sqrt{2}$. **A1**

(ii) $\sqrt{2} \le y \le 2$ **A1 A1**

[3 marks]

(b) $k = 2$ **A1**

[1 mark]

(c) $m = \sqrt{2}$

[1 mark]

(d) $-\sqrt{2} < k < \sqrt{2}$ **A1 A1**

(strict inequality) **A1**

[3 marks]

(e) $\int_0^1 \left(\sqrt{1+x} + \sqrt{1-x}\right) dx$ **M1**

$\left[\frac{2}{3}(1+x)^{\frac{3}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}}\right]_0^1$ **A1 A1**

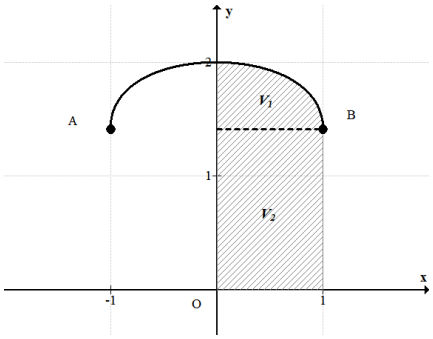
$= \frac{2}{3}(2)^{\frac{3}{2}} = \frac{4\sqrt{2}}{3}$ **A1**

[4 marks]

(f) $\pi\int_0^1\left(\sqrt{1+x}+\sqrt{1-x}\right)^2 dx = 3.57\pi$ (OR 11.2) **A1 A1**

[2 marks]

(g) We have to split the volume onto two parts. The first volume is generated by the rotation of the graph from $y=\sqrt{2}$ to $y=2$ and the second volume is a cylinder of radius 1 and height $\sqrt{2}$. **M1**



For the volume V_1:

$\sqrt{1+x}+\sqrt{1-x}=y \Leftrightarrow 1+x+1-x+2\sqrt{(1+x)(1-x)}=y^2$ **M1**

$\Leftrightarrow 2+2\sqrt{1-x^2}=y^2 \Leftrightarrow \sqrt{1-x^2}=\frac{y^2-2}{2}$

$\Leftrightarrow 1-x^2=\frac{\left(y^2-2\right)^2}{4} \Leftrightarrow x^2=1-\frac{\left(y^2-2\right)^2}{4}$ **A1**

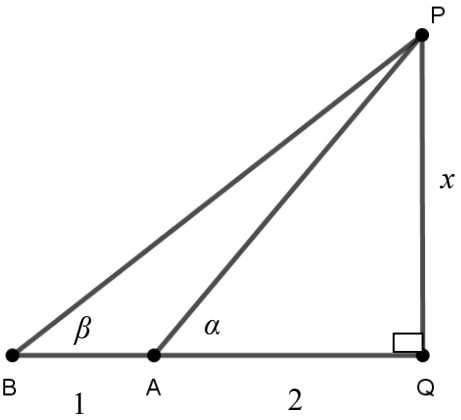
$V_1=\pi\int_{\sqrt{2}}^{2}x^2dy=\pi\int_{\sqrt{2}}^{2}1-\frac{\left(y^2-2\right)^2}{4}dy=0.4067\pi$ (= 1.27768…) **A1**

$V_2=\pi r^2h=\pi 1^2\sqrt{2}=\pi\sqrt{2}$

$V_{TOTAL}=V_2+V_2=0.4067\pi+\pi\sqrt{2}=5.72$ **A1**

[5 marks]

12. [Maximum mark: 19]



(a) $\tan a = \frac{x}{2} \Rightarrow a = \arctan\frac{x}{2}$ and similarly, $\beta = \arctan\frac{x}{3}$. **M1 A1**

[2 marks]

(b) $\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} = \frac{\frac{x}{2} - \frac{x}{3}}{1 + \frac{x}{2}\frac{x}{3}} = \frac{3x - 2x}{6 + x^2} = \frac{x}{6 + x^2}$ **M1 A1 A1**

Hence, $\alpha - \beta = \arctan\left(\frac{x}{6 + x^2}\right)$

[3 marks]

(c) $f(x) = \frac{x}{6 + x^2}$, $x > 0$

$f'(x) = \frac{(6 + x^2) - x(2x)}{(6 + x^2)^2} = \frac{6 - x^2}{(6 + x^2)^2}$ **M1 A1**

$f'(x) = 0 \Leftrightarrow 6 - x^2 = 0 \Leftrightarrow x = \sqrt{6}$ **A1**

Sign changes from + to – at this point, hence local max. **A1**

$f(\sqrt{6}) = \frac{\sqrt{6}}{12}$ **A1**

[5 marks]

(d) $f''(x) = \dfrac{-2x(6+x^2)^2 - 4x(6-x^2)\ 6+x^2}{\left(6+x^2\right)^4}$ **M1 A1**

$= \dfrac{-2x(6+x^2) - 4x(6-x^2)}{\left(6+x^2\right)^3} = \dfrac{-12x - 2x^3 - 24x + 4x^3}{\left(6+x^2\right)^3}$

$= \dfrac{2x^3 - 36x}{\left(6+x^2\right)^3}$ **A1**

$f''(x) = 0 \Leftrightarrow 2x^3 - 36x = 0 \Leftrightarrow x(x^2 - 18) = 0 \Leftrightarrow x = \sqrt{18}$

Hence, $p = \sqrt{18}\ \ (= 3\sqrt{2})$ **A1**

[4 marks]

(e) (i) $\hat{APB} = \alpha - \beta$ and its maximum value is $\arctan\dfrac{\sqrt{6}}{12} = 11.5°$ **M1 A1**

(ii) $\dfrac{x}{6+x^2} = \tan 10°$ **M1**

$x \cong 1.41$ or $x \cong 4.26$ **A1 A1**

[5 marks]